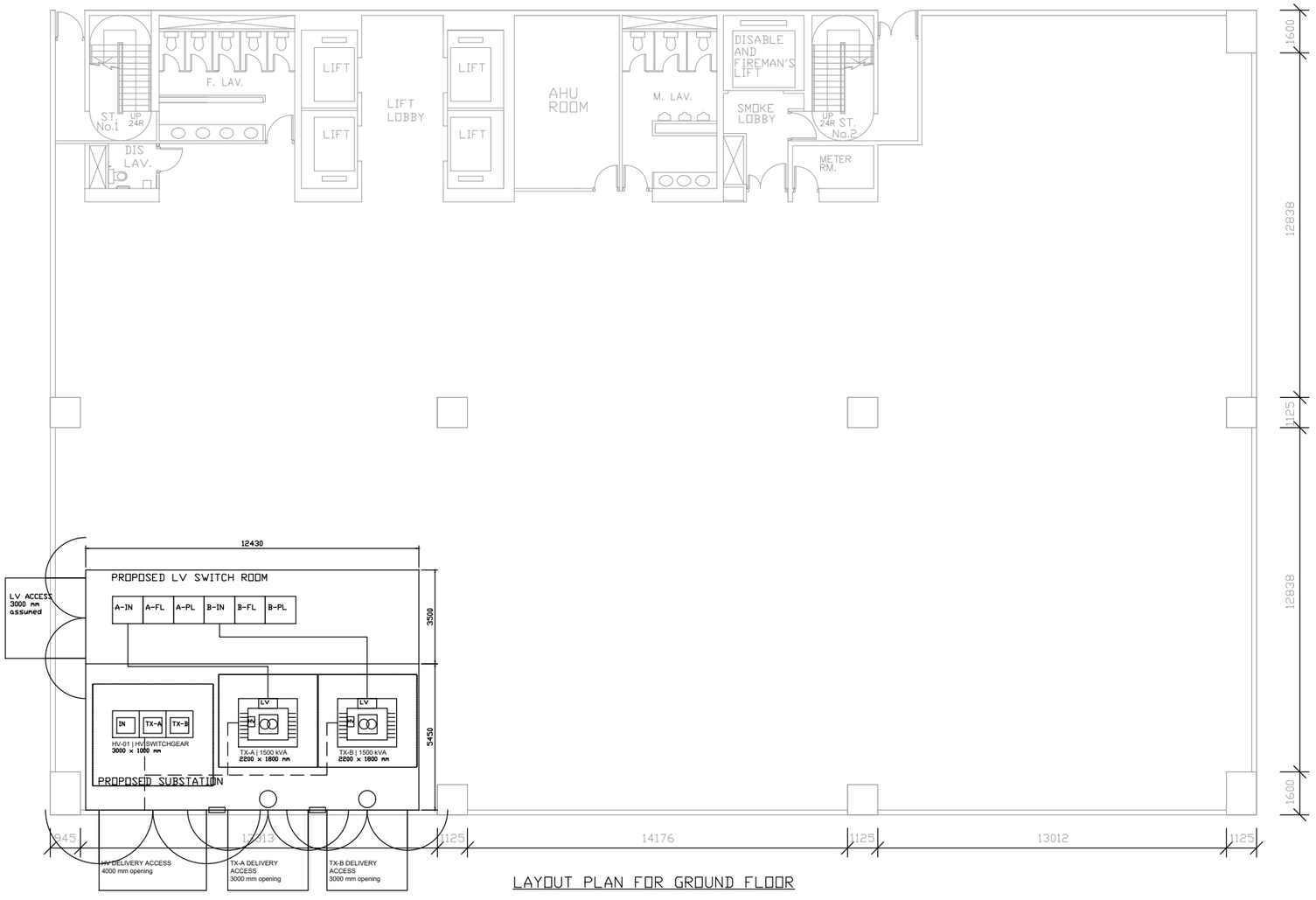
TASK 1 | GROUND-FLOOR PLANTROOM LOCATION

A3 LANDSCAPE | SCALE 1:200 | DIMENSIONS IN mm | GROUND FLOOR



LAYOUT PLAN FOR GROUND FLOOR

PROPOSED LOCATION
Lower-left perimeter of supplied ground-floor plan.
Substation 12430 x 5450; adjacent LV room 12430 x 3500 assumed. See separate 1:75 room detail.

LOCATION REASONS
Direct external plant access is proposed, subject to site confirmation.
Adjacent LV room keeps transformer-to-incomer interfaces short.
Perimeter plantrooms separate supply equipment from the upper lift/stair/AHU core and remaining accommodation.
LV room has its own proposed west entrance.

LOAD ASSESSMENT
Offices: 1644.800 kVA.
AC: 1000 kW cooling -> 526.039 kW electrical -> 584.488 kVA at assumed PF 0.90.
Public services: 200 kVA aggregate allowance.
Two-zone conservative envelope: 2672.216 kVA.
Propose 2 x 1500 kVA, with equal zone allocation; no full-load N-1.

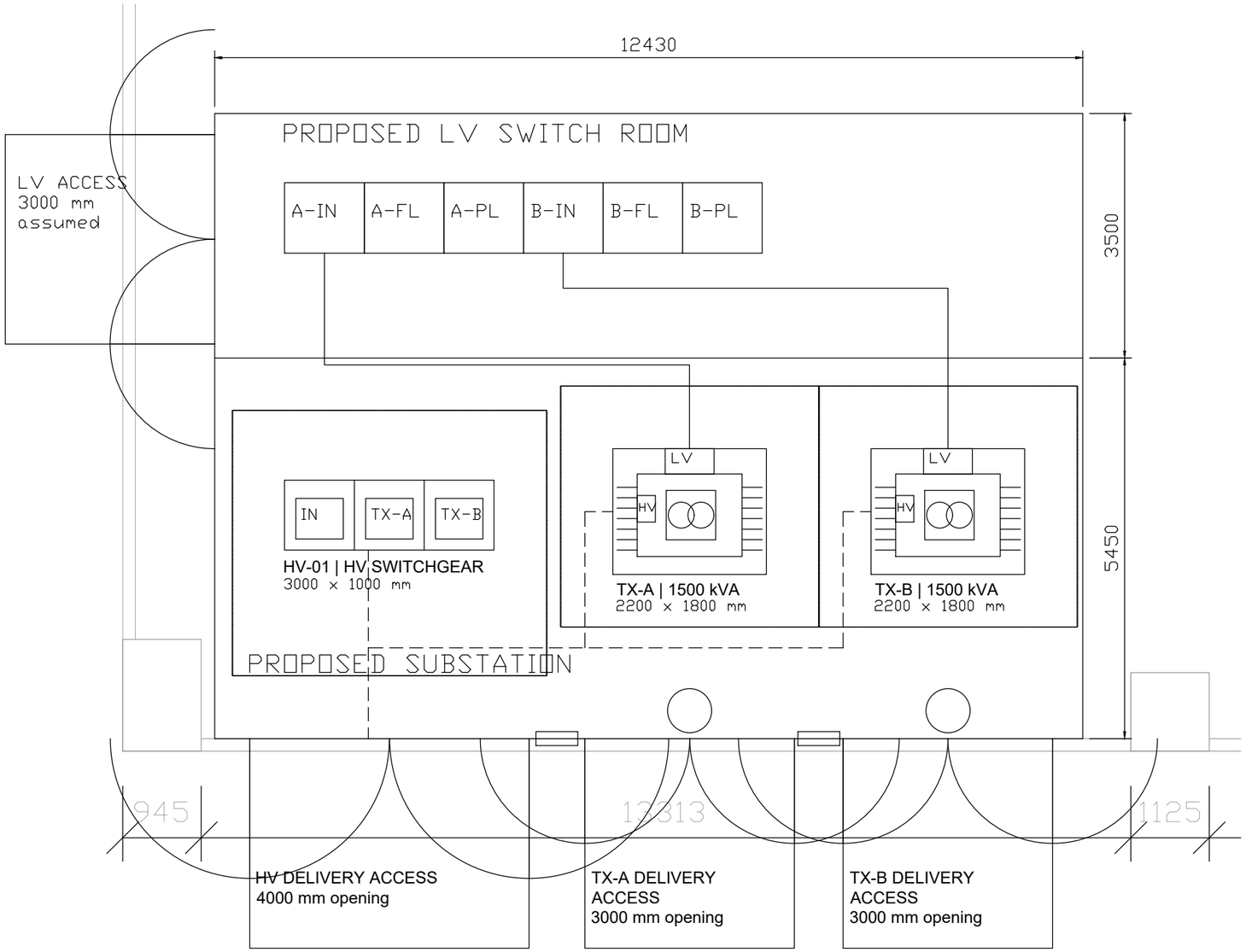
SITE CONDITIONS TO VERIFY
External apron, road and <=20m unloading route are not given by the source plan. Maintain >=3m clear delivery route.
Architectural wall openings, structural/flood/fire provisions, ventilation and supplier equipment approval require coordination.

GROUP 1 | TASK 1

Design basis: CLP COP215 Rev07; CLP COP101 v16.

TASK 1 | PROPOSED SUBSTATION AND MAIN LV ROOM

A3 LANDSCAPE | SCALE 1:75 | DIMENSIONS IN mm | PLANTROOM DETAIL



LINE KEY: dashed = HV trench; solid = LV route; chain = clearance. Equipment details are diagrammatic.

LV INTERFACES / HV ROUTES
LV trench: 1000 W x 1100D, four MCTs per interface. A: 8.03+2=10.03m; B: 8.31+2=10.31m (<20m). 2m is assumed total termination/vertical allowance.
HV centre-lines: 1000 W at panels; 600 W branches; final 300 W; generally 1400D. Segregate HV/LV. Covered trenches; no beam obstruction.

DETAIL 1:75 AT A3

CAPACITY AND EQUIPMENT
TX-A / TX-B: 1500 kVA each. Total 3000 kVA; design load 2672.216 kVA. No full-load N-1.
HV-01: 3000 x 1000 mm; TX-A/B: 2200 x 1800 mm.
Cabinets: A/B-IN incomer; A/B-FL floor feeders; A/B-PL plant/public feeders.
Generic plan symbols; overall dimensions are adopted design assumptions. Internal details are diagrammatic.

CLEARANCES / BUILDING WORKS
HV rear 1000, front 1800, sides 750. TX LV terminals north: 900; other sides 750.
Proposed 2h FRR partitions; fire-seal penetrations. FFL +0.15 vs external +0.00 assumed.
Actual structures, wall thickness and equipment dimensions require coordination.

ACCESS AND DOORS
South external access assumed. Provide 3000-wide clear delivery route; unloading distance <=20 m to verify.
Plant doors: HV 4000 W; TX 3000 W each; 2800 H assumed. Outward opening, shown at 90 degrees.
Adjacent plant doors used sequentially; TX-A has wicket personnel access.
Independent LV entrance west.

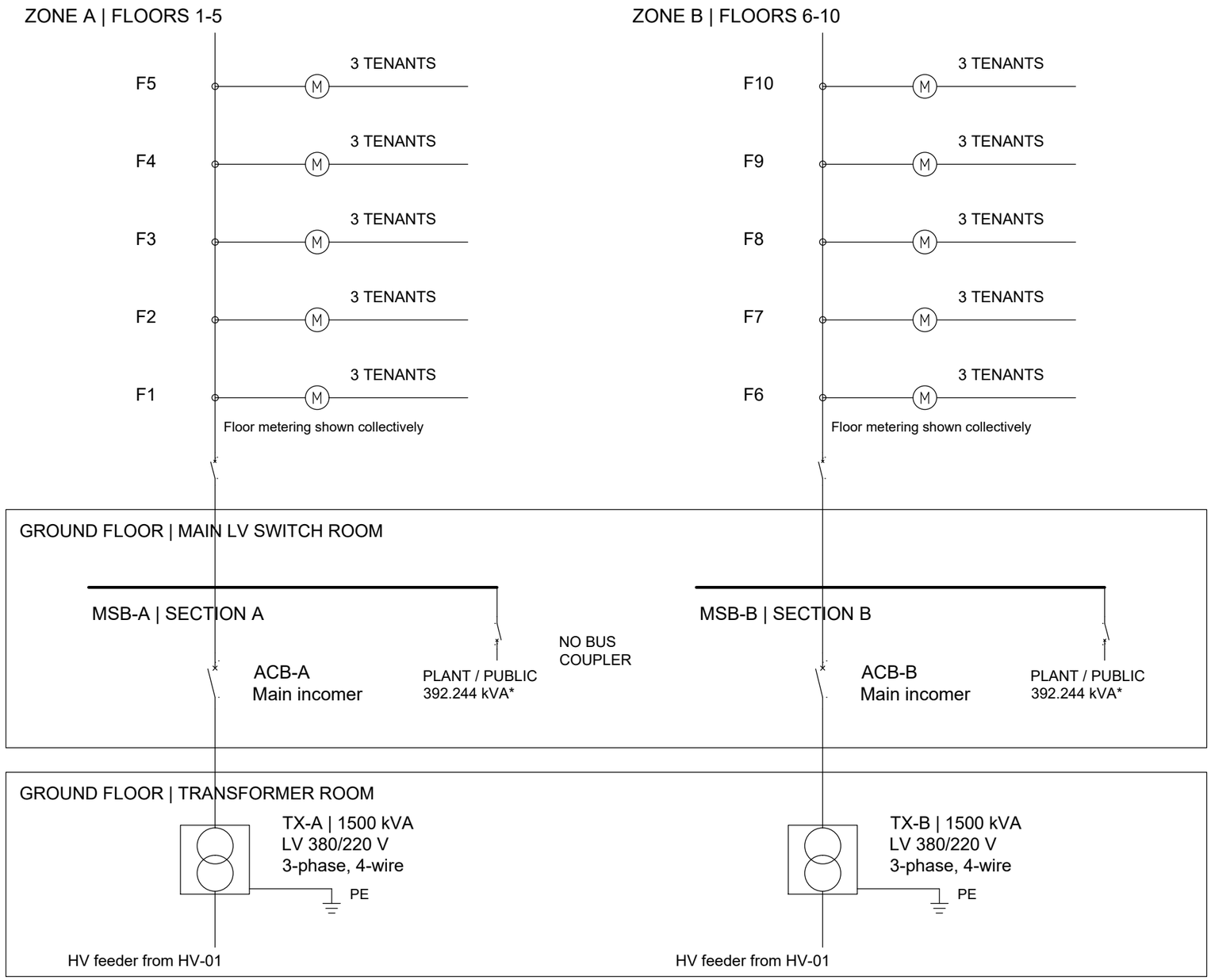
VENTILATION
2 low-level 600 W x 2000H inlet louvres: 2.4m2 gross x assumed 50% free area = 1.20m2 >=1.12m2. Certified free area >=46.67% required.
2 high-level 630-dia fan symbols are conditional references. Actual losses, fan duty and noise to confirm; outlet >=2.5m above street, away from people.

Basis: supplied COP215 Rev07; COP101 v16.
See the new main distribution schematic for TX-A/B, MSB-A/B and floor supply zones.

TASK 1 | MAIN ELECTRICAL DISTRIBUTION SCHEMATIC

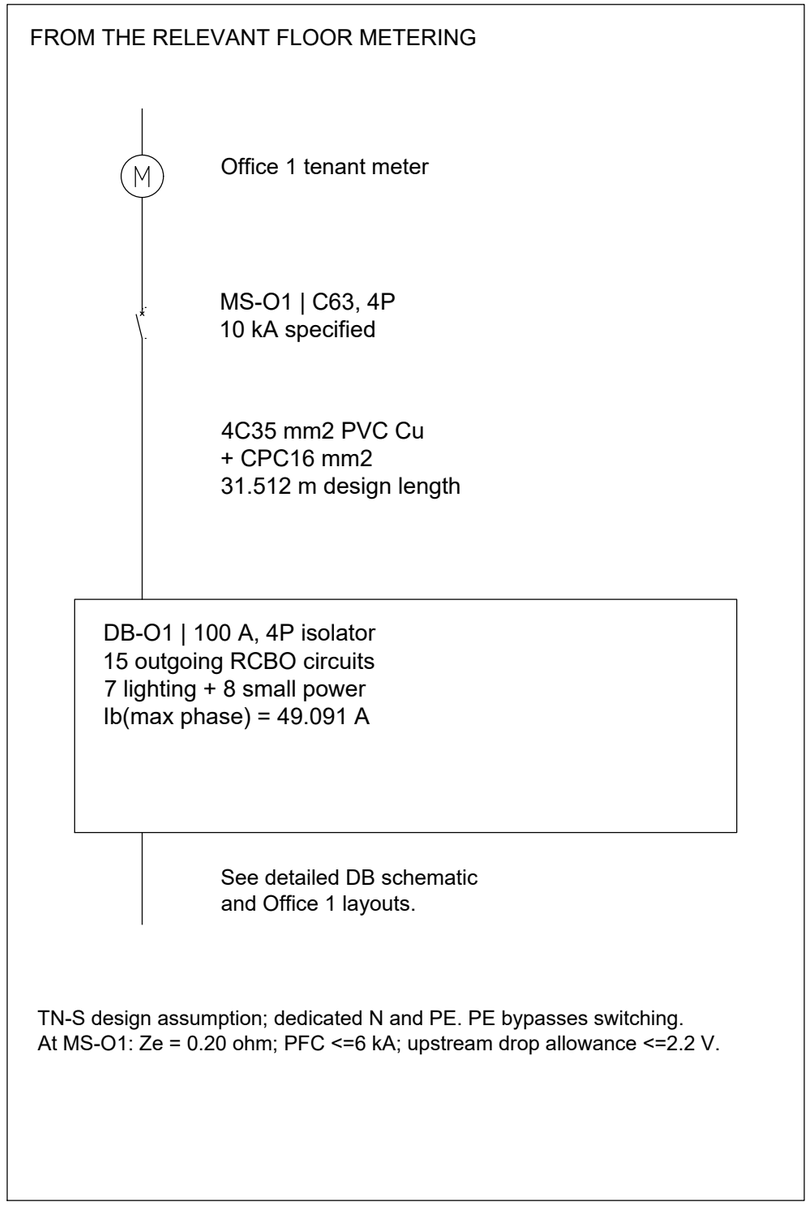
A3 LANDSCAPE | NOT TO SCALE | 2 x 1500 kVA | 380/220 V LV DESIGN BASIS

NORMAL SUPPLY: 2672.216 kVA total; two independent LV sections.
Equal allocation: 1336.108 kVA / approx. 2030 A per section.



* Per section: AC 292.244 + public services 100 = 392.244 kVA before the 1.1 zoning factor.

TYPICAL OFFICE 1 SUPPLY
Selected tenant detail; applicable to its floor supply zone.



TN-S design assumption; dedicated N and PE. PE bypasses switching.
At MS-O1: Ze = 0.20 ohm; PFC <=6 kA; upstream drop allowance <=2.2 V.

SYSTEM BASIS
HV supply voltage and upstream protection: utility interface. Capacities and LV voltage follow Task 1. Main ACB ratings/settings and fault duties require coordinated equipment selection. PE denotes equipment bonding; source earthing arrangement to be coordinated.

DESIGN LIMITS
No full-load N-1. Central plant/public load is split equally as a design assumption. This sheet shows normal supply; emergency systems are separate.

REFERENCE: Electrical Schematic 1 - transformer, main switchboard and floor-metering arrangement.
Read with the plantroom layout and Office 1 distribution-board schematic.

GROUP 1 | E-04 | REV 01 | 10 NOV 2026